

# Merck Capstone Project

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# Metadata Analysis

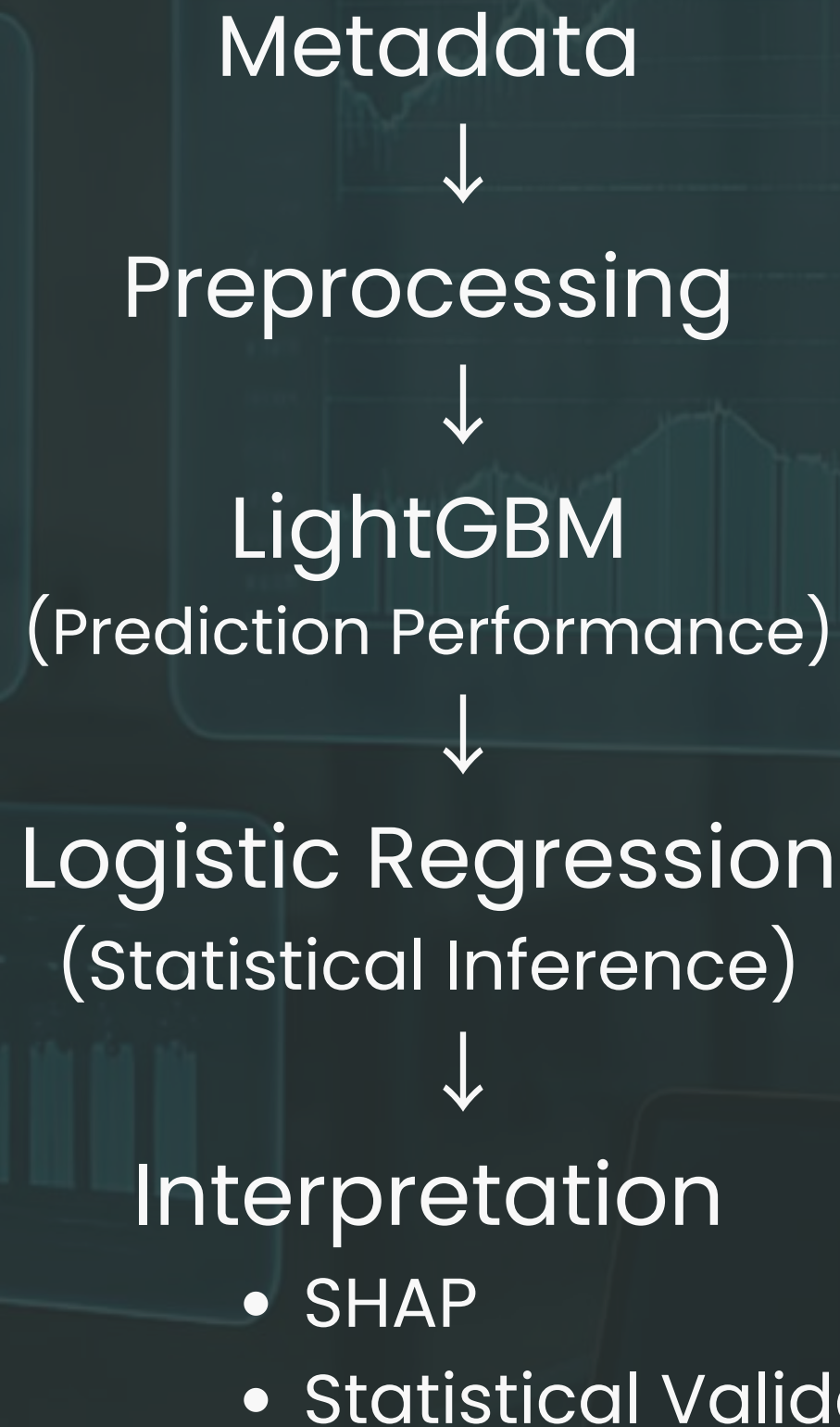
## Research Question

Can demographic, behavioural and MRI metadata be used to predict ADHD diagnosis and biological sex?

### Metadata:

- Demographics
- SDQ
- APQ
- EHQ
- MRI Acquisition Metadata

# Metadata Analysis Workflow



## Preprocessing

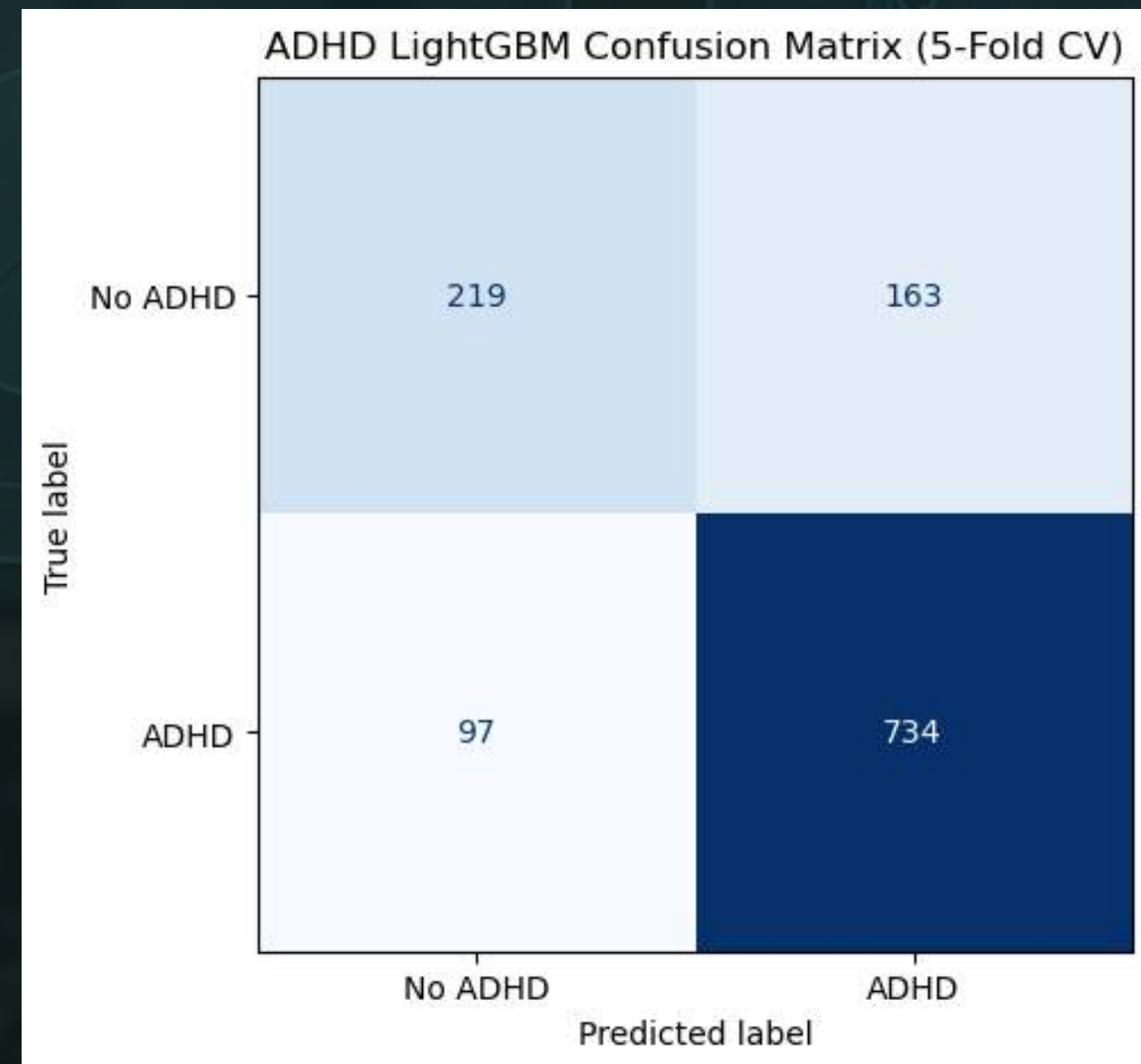
- Iterative Imputation
- One-Hot Encoding
- Ordinal Encoding (Barratt variables)

## Model Setup

1213 Participants  
38 Predictors  
ADHD & Sex Prediction  
Stratified 5-Fold CV

# ADHD Prediction

## Model Performance

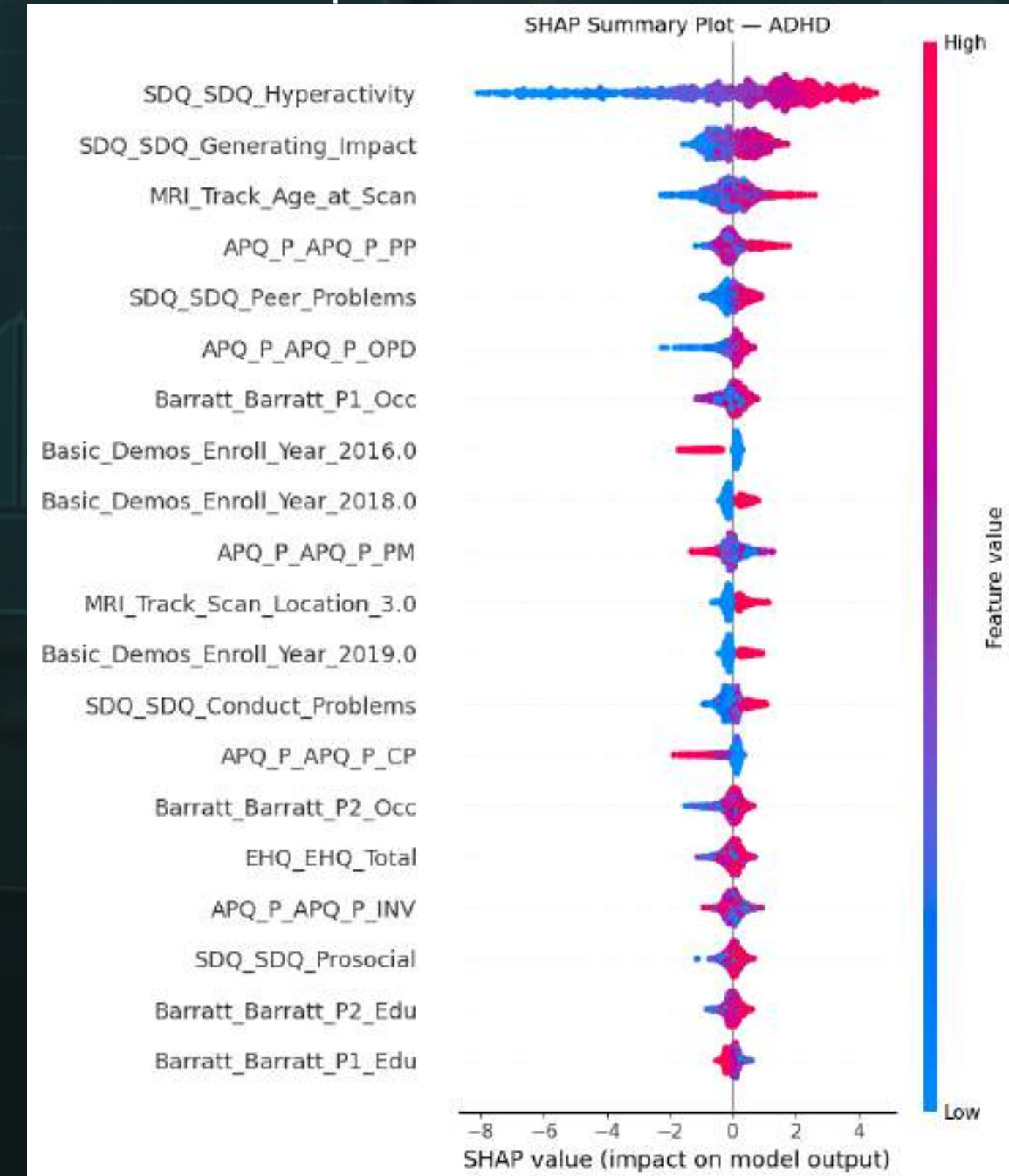


Accuracy  
78.6%

ROC-AUC  
0.810

Balanced Accuracy  
72.8%

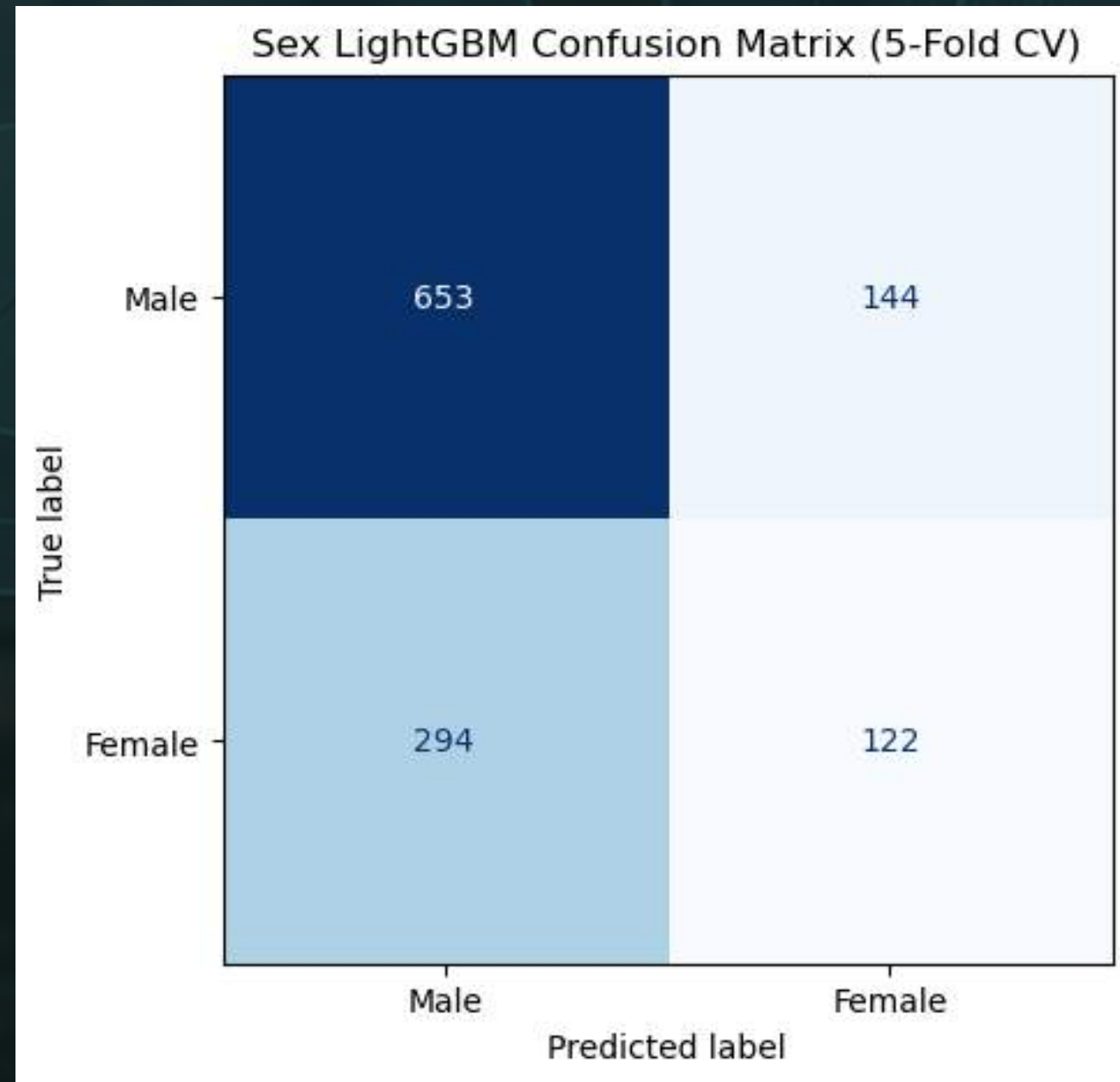
## Feature Importance (SHAP)



Behavioural variables, particularly SDQ Hyperactivity, were the strongest contributors to ADHD prediction.

# Sex Prediction

## Model Performance

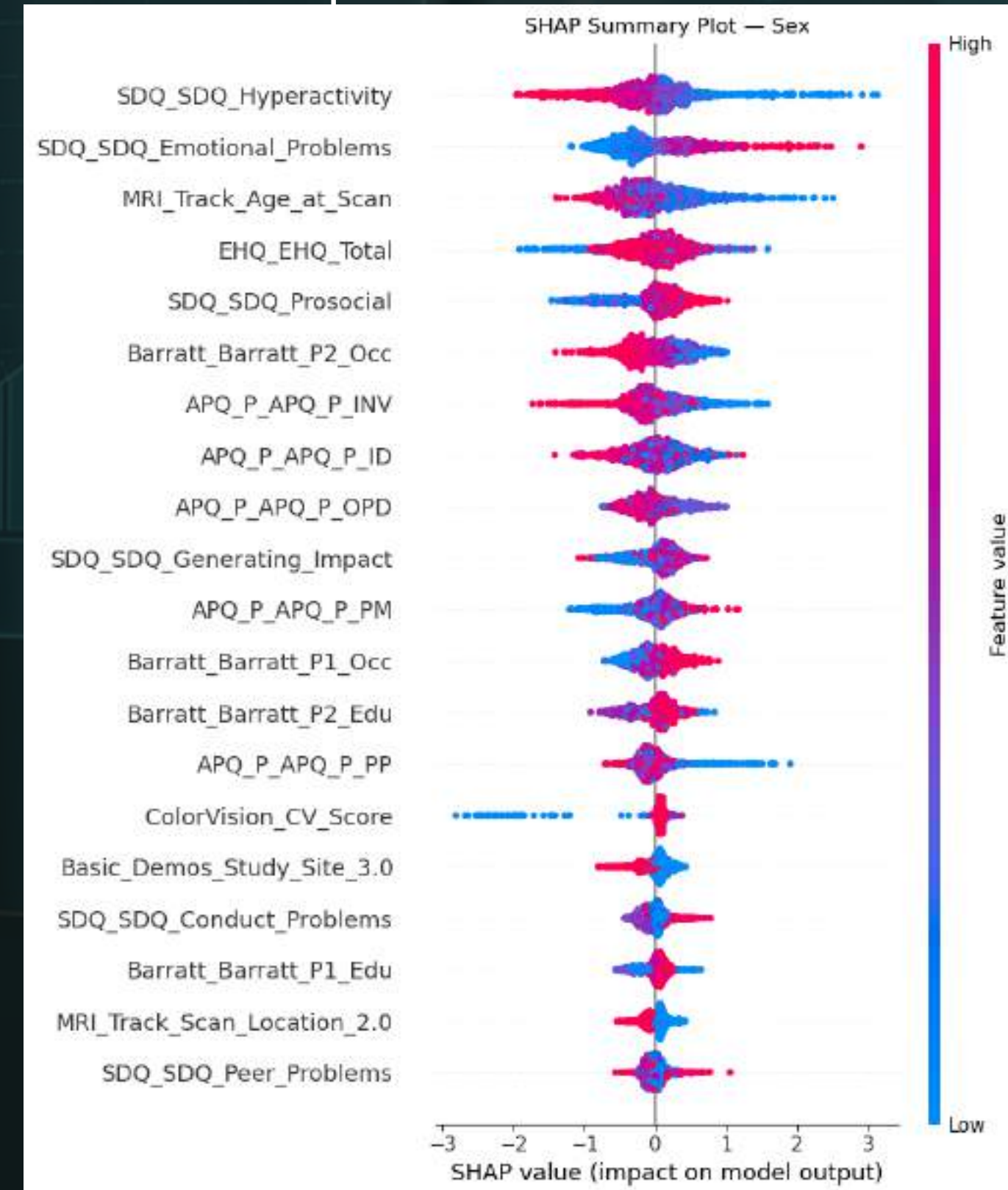


Accuracy  
63.5%

ROC-AUC  
0.52

Balanced Accuracy  
49.6%

## Feature Importance (SHAP)



Metadata alone showed limited ability to accurately predict biological sex.

# Logistic Regression & Model Diagnostic

|                         | ADHD            | Sex             |
|-------------------------|-----------------|-----------------|
| Converged               | ✓               | ✓               |
| VIF                     | <2.2            | <2.2            |
| Box-Tidwell             | Passed          | Passed          |
| Overall Model           | Not Significant | Not Significant |
| LR Test                 | p = 0.784       | p = 0.501       |
| McFadden R <sup>2</sup> | 0.021 (Low)     | 0.024 (Low)     |

Model assumptions were satisfied, but logistic regression indicated that metadata alone explained only a small proportion of outcome variability.

# Conclusions

## ADHD

- Reasonable predictive performance (AUC = 0.81)
- SDQ Hyperactivity most important feature
- Behavioural questionnaires contributed most to prediction

## Sex

- Weak predictive performance (Balanced Accuracy = 49.6)
- Metadata alone was insufficient for predicting biological sex
- Additional information (e.g. imaging features) may be required

## Clinical Interpretation

Behavioural metadata effectively predicted ADHD, whereas metadata alone had limited ability to predict biological sex.

# **fMRI Analysis**

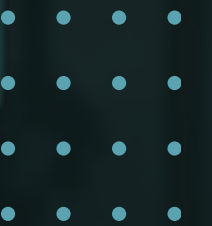
## **Research Question**

Can fMRI data predict ADHD diagnosis and biological sex?

- Evaluate the predictive ability of resting-state fMRI connectivity.
- Compare prediction performance for Sex\_F and ADHD\_Outcome.
- Identify reproducible connectivity features associated with prediction.

# fMRI Data Description

- Functional Magnetic Resonance Imaging (fMRI) is a neuroimaging technique that tracks brain activity over time (unlike structural MRI, which captures static anatomy).
- The BOLD Signal: It measures the Blood-Oxygen-Level-Dependent contrast, an indirect view of neural activity. When brain regions become active, local blood flow increases, changing the ratio of oxygenated to deoxygenated blood.
- From Biology to Features:
  - The scanner records these fluctuations as a time-series.
  - By measuring how synchronously different brain regions fluctuate (functional connectivity), we build the matrices used to train models for identifying ADHD and sex-specific neural patterns.



# Modelling

Raw fMRI Connectivity Data (1,213 participants, 19,900 features)



Fisher's z-Transformation



5 Fold Stratified Cross Validation



L1 - Logistic Regression

L1 Regularization Automatic Feature Selection

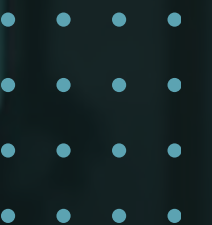
XGBoost

Welch's t-test Top 1000 Features

Performance Evaluation: Accuracy • Balanced Accuracy • F1 • ROC-AUC



Compare Model Performance



# Results

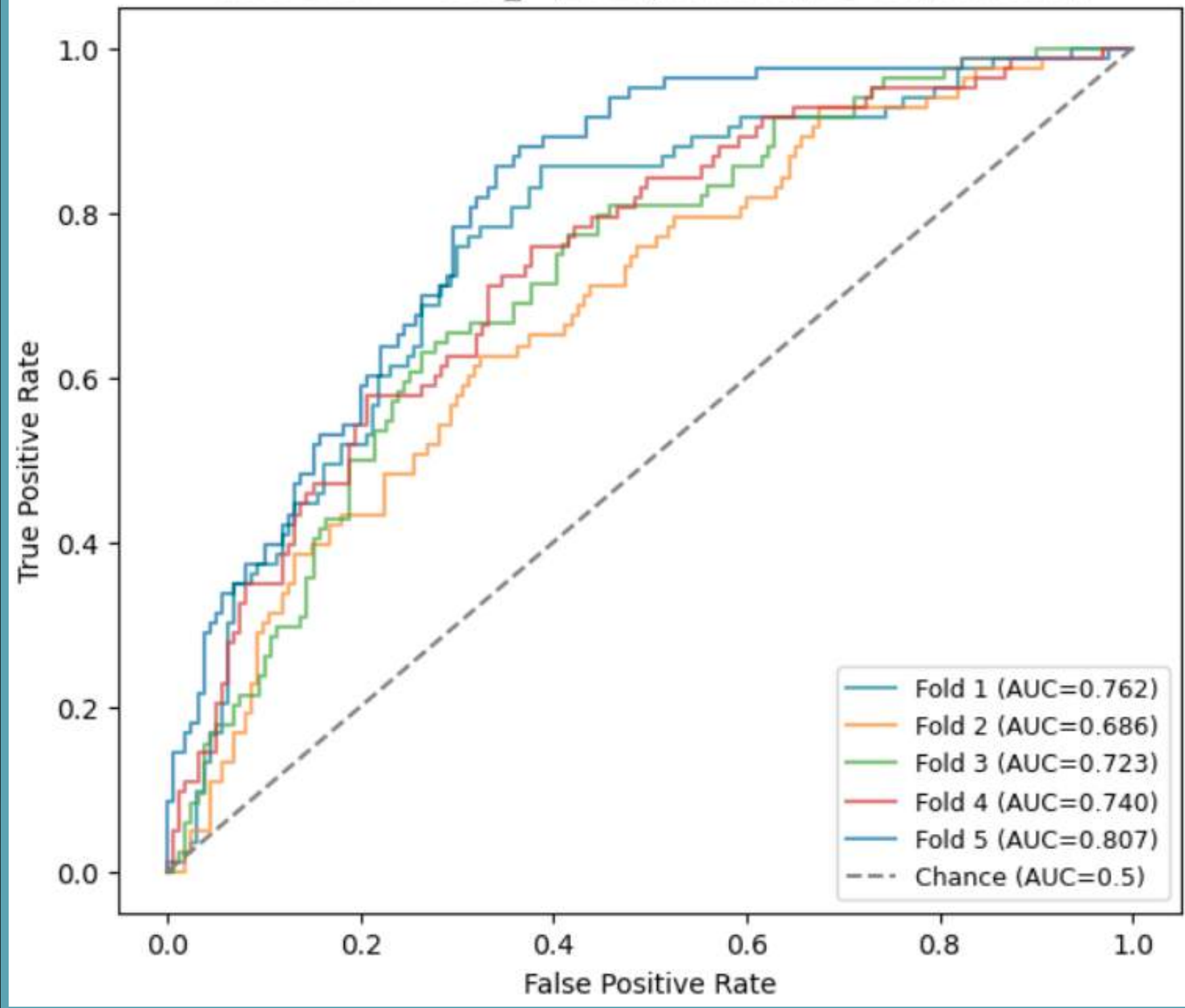
| L1 Regularized Logistic Regression |       |              |
|------------------------------------|-------|--------------|
| Metric                             | Sex_F | ADHD Outcome |
| Accuracy                           | 0.702 | 0.580        |
| Balanced Accuracy                  | 0.666 | 0.537        |
| F1-Score                           | 0.555 | 0.678        |
| ROC-AUC                            | 0.743 | 0.553        |
|                                    |       |              |

| XGBoost           |       |              |
|-------------------|-------|--------------|
| Metric            | Sex_F | ADHD Outcome |
| Accuracy          | 0.674 | 0.645        |
| Balanced Accuracy | 0.599 | 0.530        |
| F1-Score          | 0.429 | 0.764        |
| ROC-AUC           | 0.695 | 0.554        |
|                   |       |              |

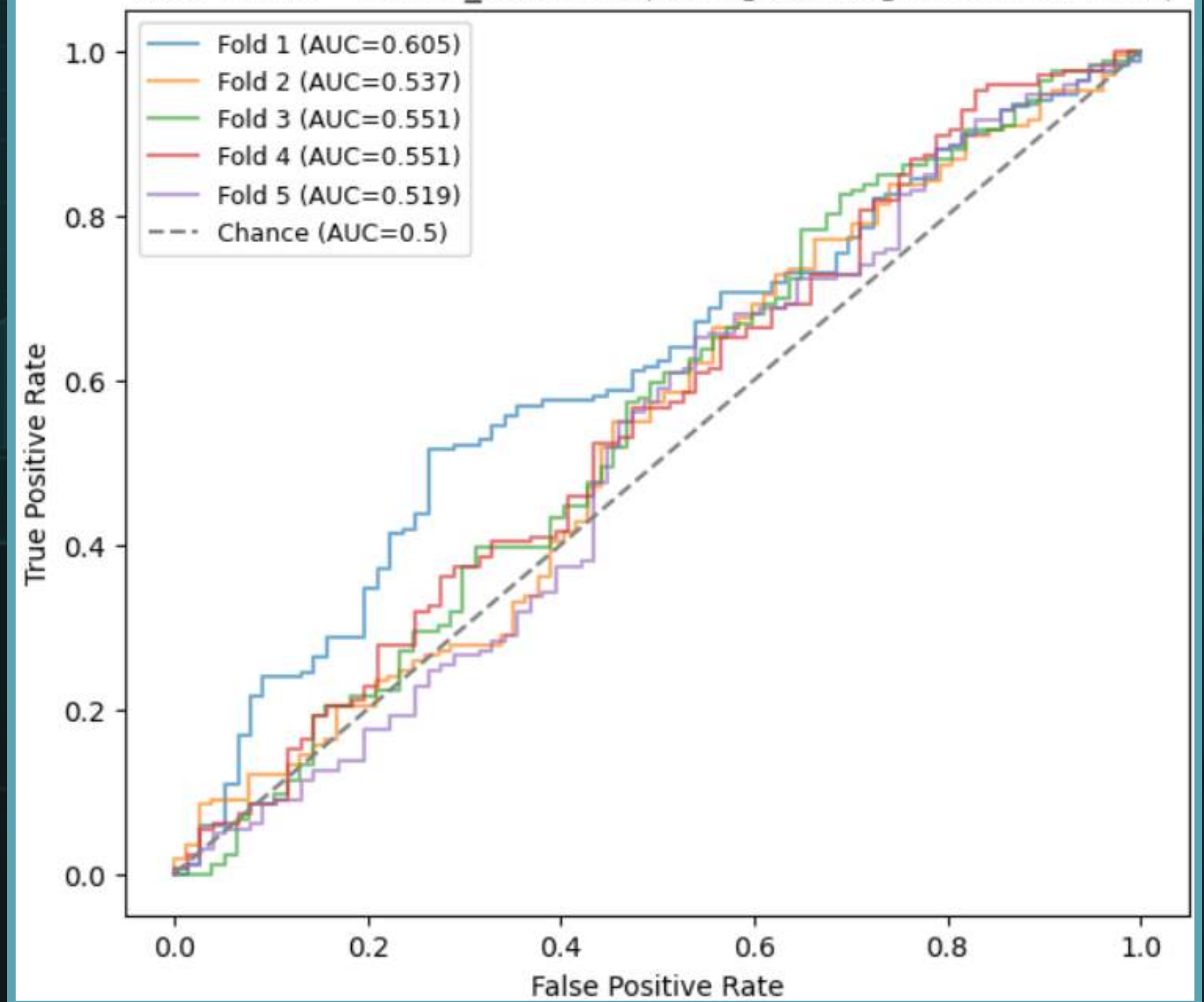
# ROC Curve Comparison

## L1 Logistic Regression

ROC Curve — Sex\_F (L1 Logistic Regression, all folds)



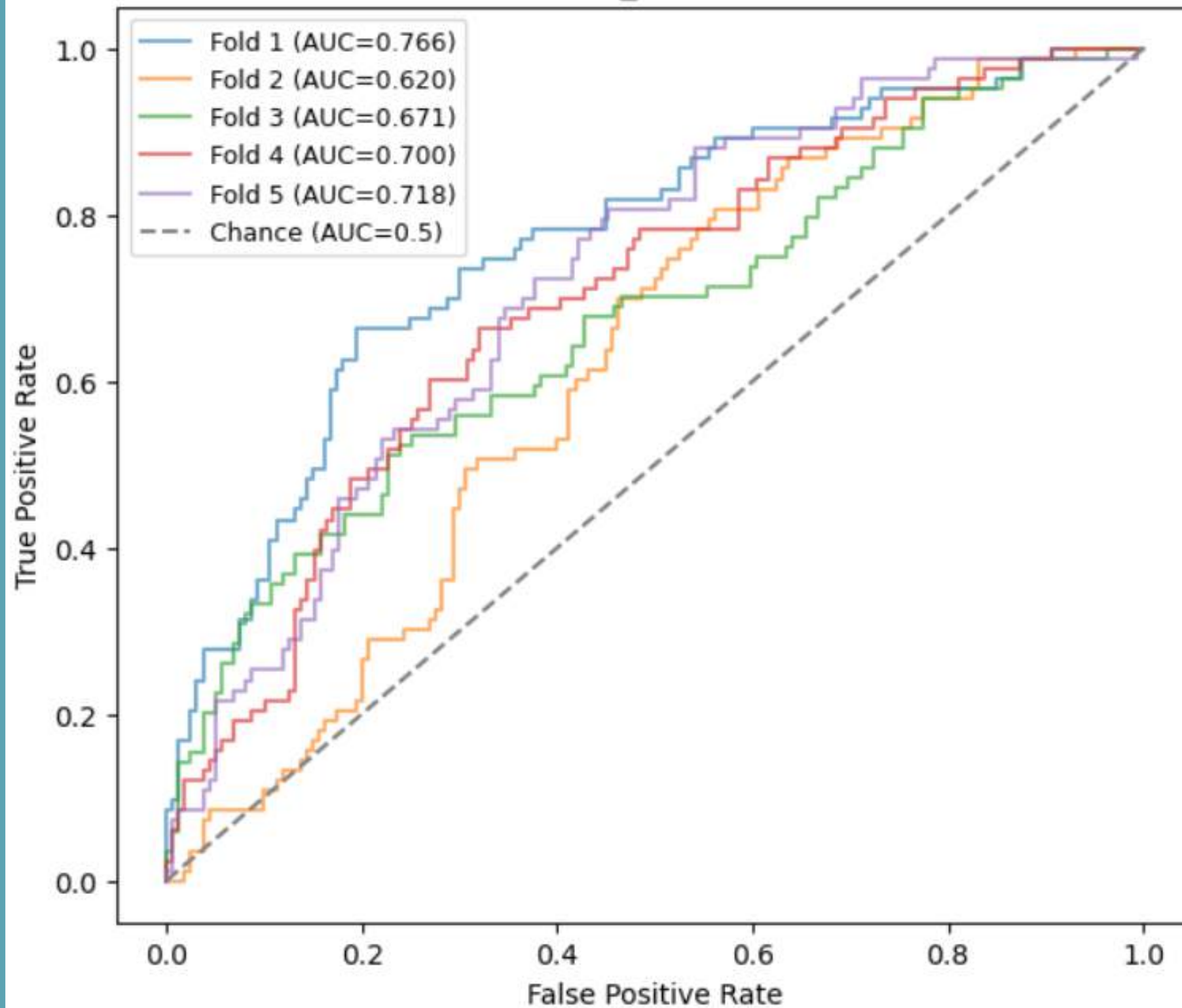
ROC Curve — ADHD\_Outcome (L1 Logistic Regression, all folds)



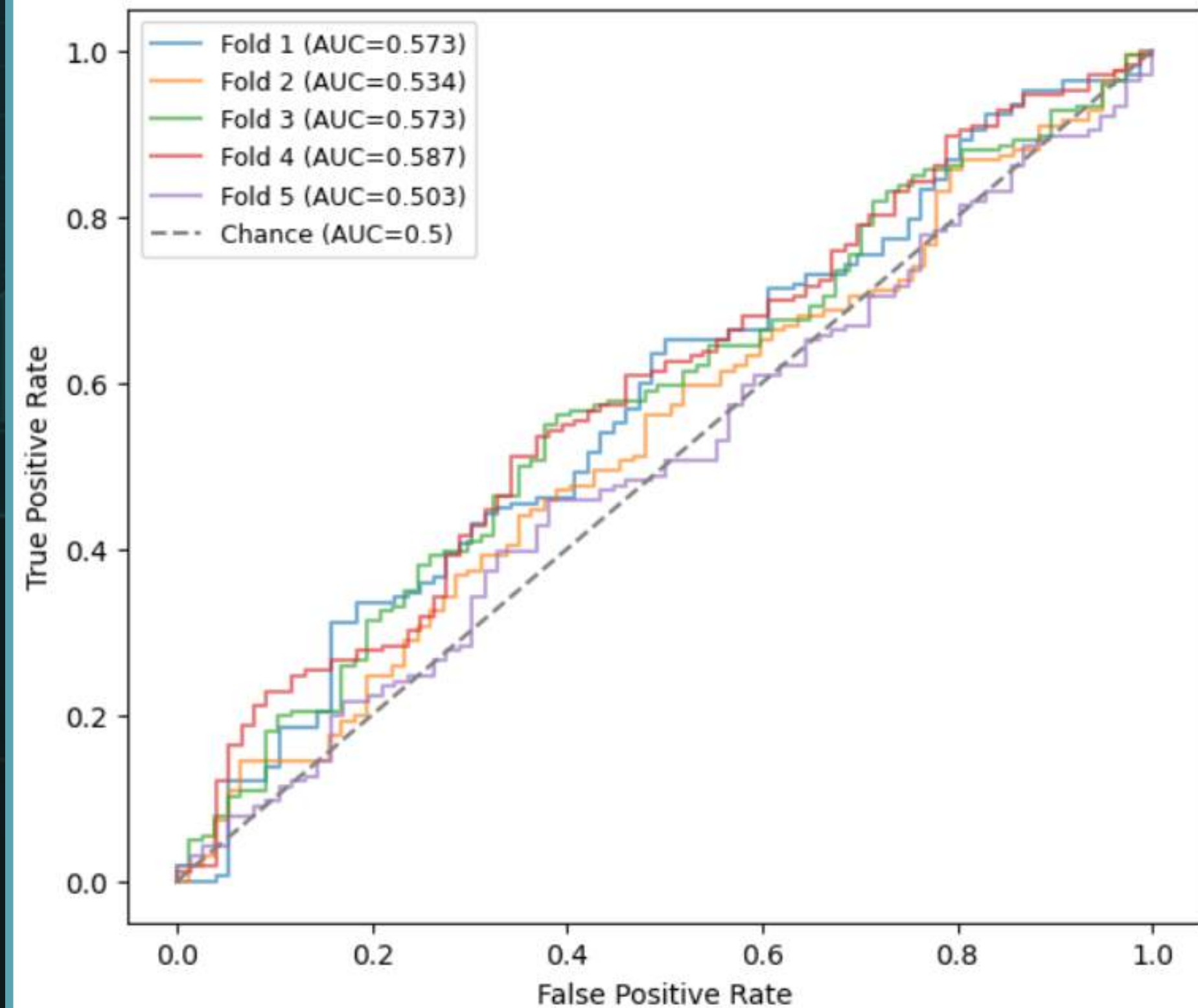
# ROC Curve Comparison

## XGBoost

ROC Curve — Sex\_F (XGBoost, all folds)



ROC Curve — XGBoost, all folds



# Conclusion

## PREDICTING SEX\_F

- ROC-AUC = 0.743
- Reliable Signal
- L1 > XGBoost

## PREDICTING ADHD\_OUTCOME

- ROC-AUC ~ 0.55
- Near chance
- Similar performance by both models

343 stable connectivity features were consistently identified across cross validation folds.



# Multimodal Machine Learning for ADHD and Biological Sex Prediction

## Research Question:

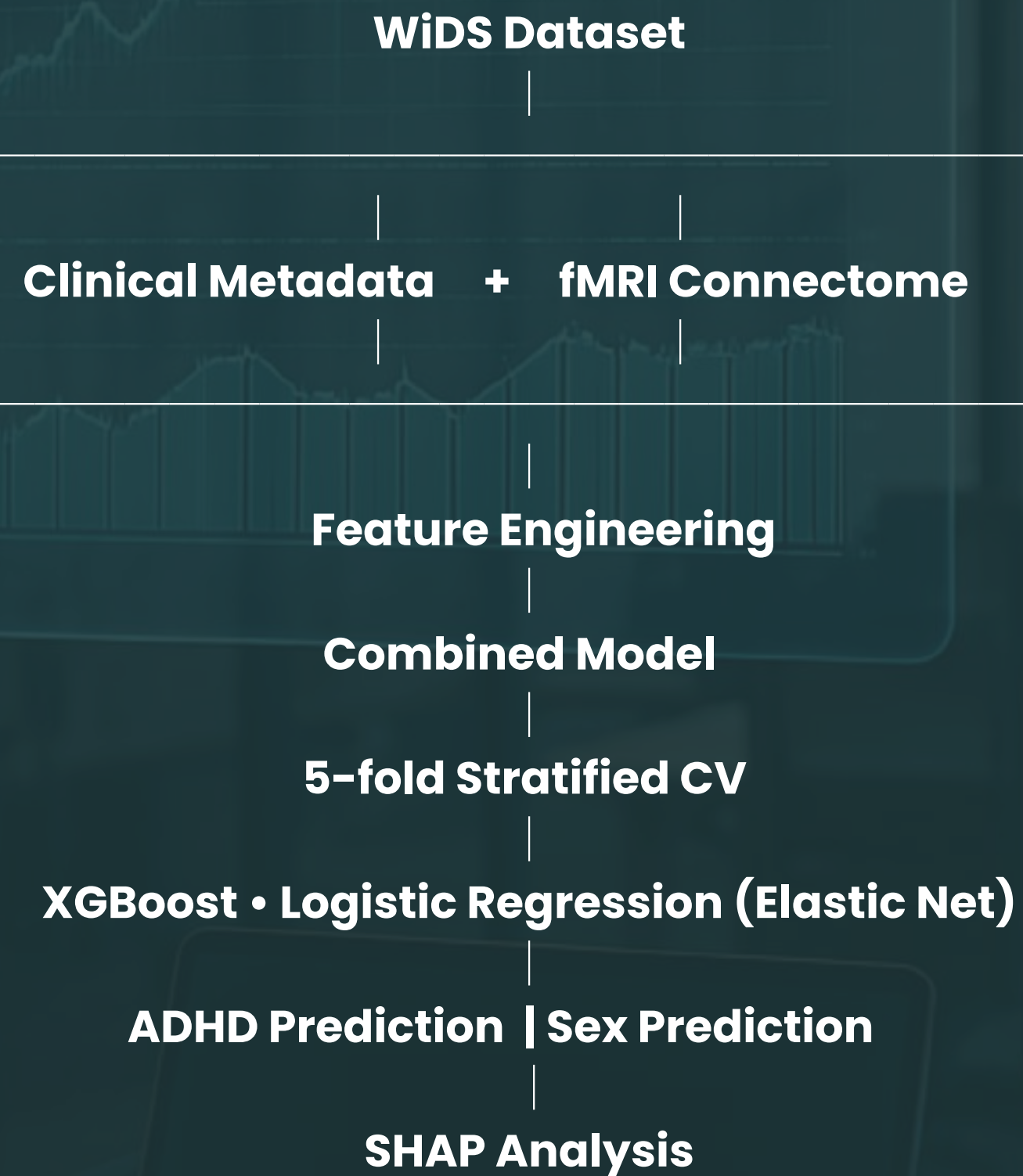
Does combining fMRI connectivity with clinical metadata improve prediction of both ADHD and biological sex?

- Evaluate model performance for ADHD prediction and Biological Sex classification with Combined model.
- Compare predictive performance of Clinical Metadata, fMRI Connectivity, and Combined models.



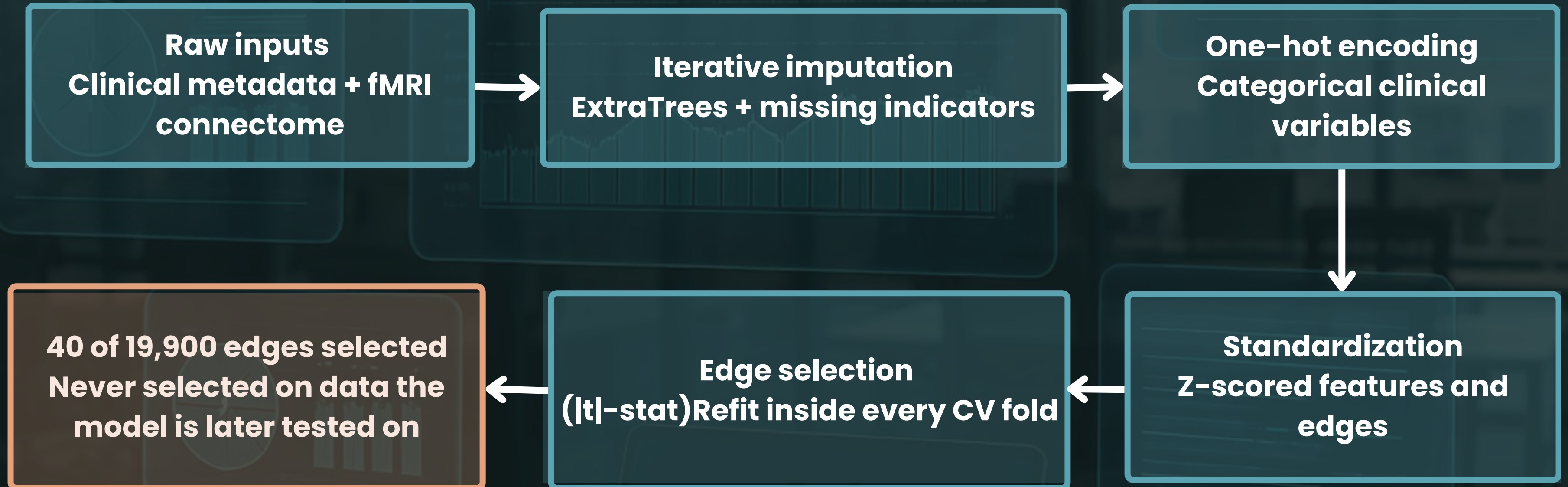


# Multimodal Machine Learning Pipeline





# Feature Engineering for Combined Model





# Combined Model — What It Is : ADHD vs Sex

## ADHD - Combined Model Prediction

| Metric            | XGBoost | Logistic Regression |
|-------------------|---------|---------------------|
| AUC               | 0.82    | 0.8                 |
| Accuracy          | 0.78    | 0.73                |
| Balanced Accuracy | 0.73    | 0.71                |
| F1 Score          | 0.84    | 0.8                 |

## Sex - Combined Model Prediction

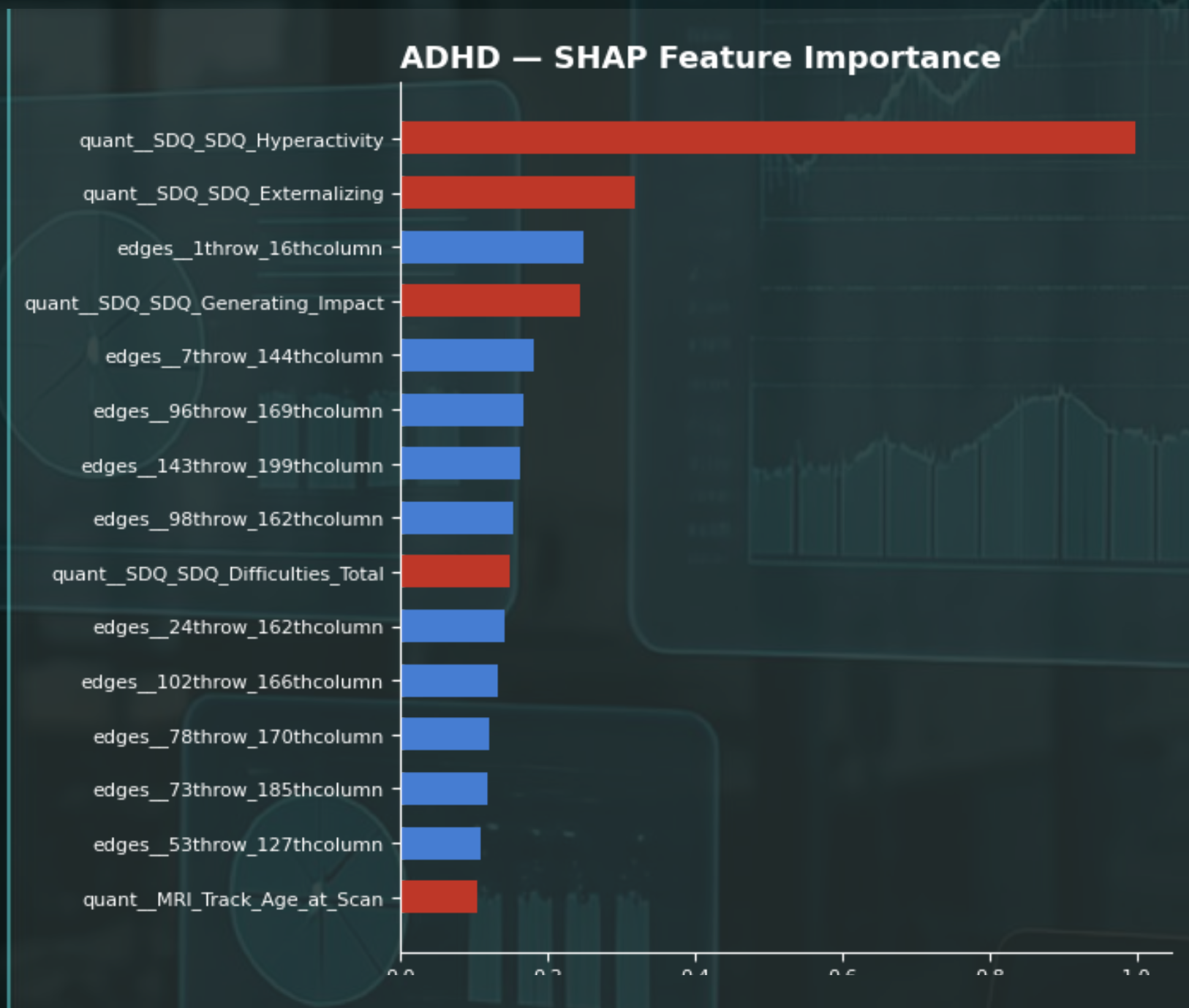
| Metric            | XGBoost | Logistic Regression |
|-------------------|---------|---------------------|
| AUC               | 0.67    | 0.67                |
| Accuracy          | 0.66    | 0.64                |
| Balanced Accuracy | 0.6     | 0.62                |
| F1 Score          | 0.46    | 0.51                |

### Key observations:

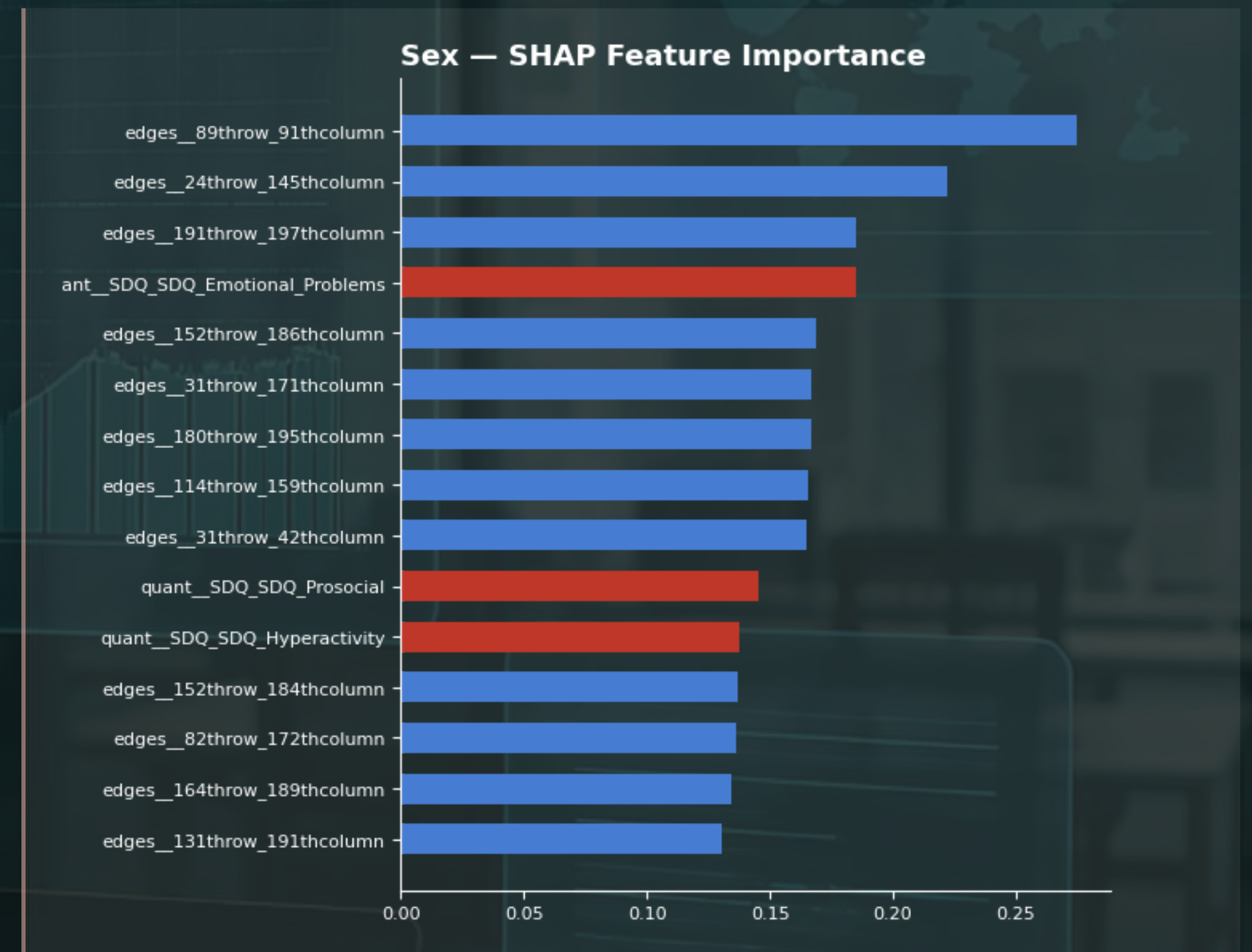
- ADHD: XGBoost outperforms Logistic Regression across all four metrics
- Sex: No clear winner — XGBoost leads on AUC, Logistic Regression on the rest
- Overall: Sex prediction underperforms ADHD prediction across every metric

# SHAP FEATURE IMPORTANCE

● Metadata features ● fMRI connectome edges



**Metadata features dominate — behavioral scores outrank nearly all connectome features.**



**fMRI edges dominate — connectome features outrank most metadata, matching the edges AUC improvement seen earlier**

# Model Performance Comparison (ROC–AUC)- ADHD vs Sex

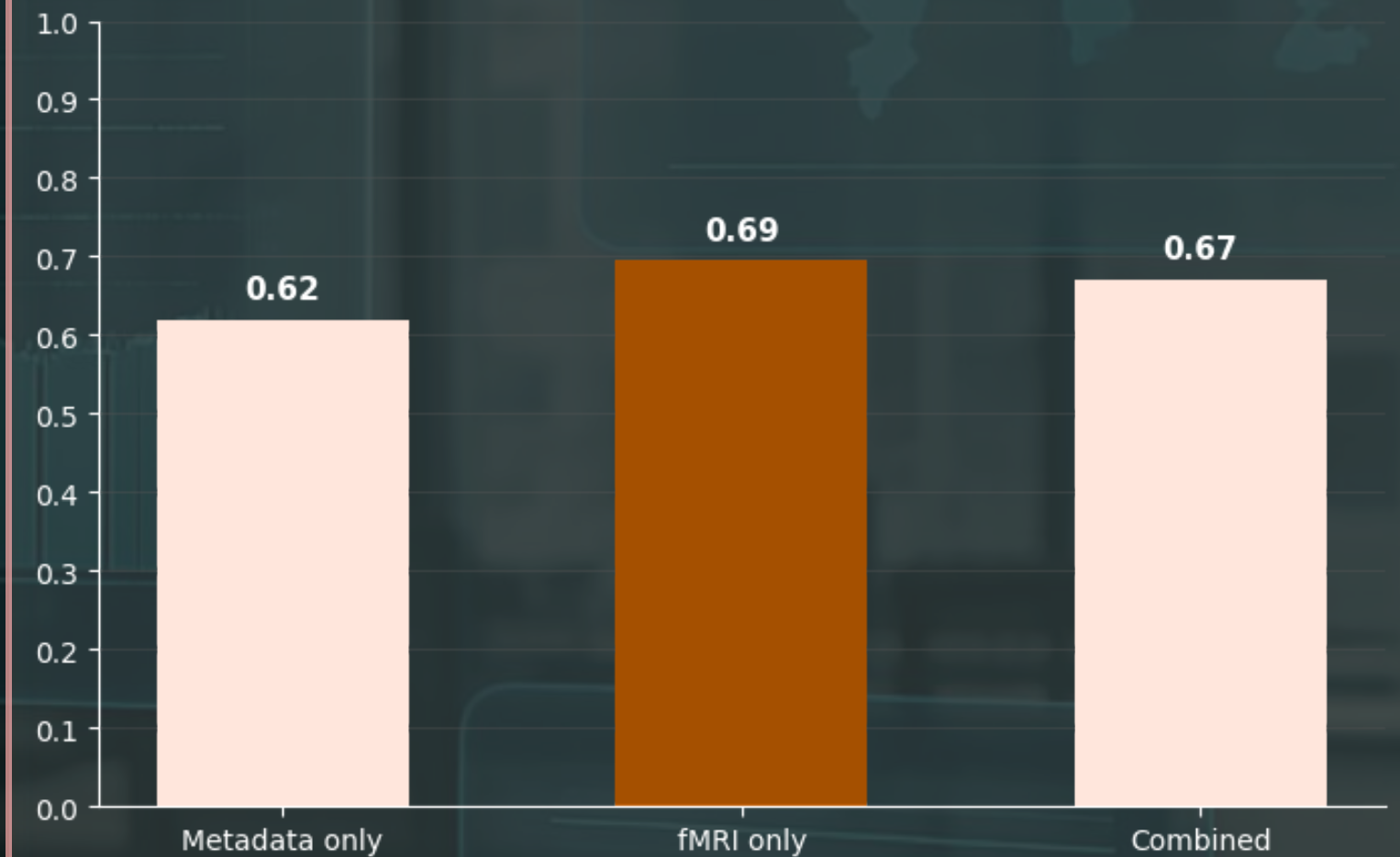
**ADHD Prediction: AUC by Feature Set**  
Metadata carries the signal



## Key Findings

- Metadata consistently achieved the highest AUC across all models
- fMRI-only models performed close to chance level (AUC  $\approx 0.55$ )
- Multimodal integration did not improve ADHD prediction

**Sex Prediction: AUC by Feature Set**  
fMRI connectivity adds signal



## Key Findings

- fMRI alone achieved the highest predictive performance (AUC 0.69)
- fMRI contributed greater predictive information than metadata alone
- Combining metadata and fMRI did not further improve performance, and slightly underperformed fMRI alone

**For ADHD, metadata alone is sufficient — fMRI adds nothing. For sex, fMRI alone is actually the strongest single signal...**



# So, Does It Help? – Conclusion

Does combining fMRI connectivity with clinical metadata improve prediction of both ADHD and biological sex?

→ No – the answer differs by ADHD and Sex.



## For ADHD

- **Clinical metadata alone is sufficient (AUC 0.83)**
- **fMRI connectivity adds no value – performance did not improve, and combining significantly hurt two of three models**
- **SHAP confirms this: behavioral scores (Hyperactivity, Externalizing) dominate feature importance, not connectome edges**

## For Biological Sex

- **fMRI connectivity alone is the strongest predictor (AUC 0.69), outperforming metadata alone (0.62)**
- **Combining does not add further improvement**
- **SHAP confirms this: connectome edges dominate feature importance, particularly default-mode network connections**

**Metadata predicts ADHD, connectivity predicts sex – but combining fMRI connectivity with clinical metadata improves neither.**

# **Are the predictors and functional-connectivity patterns associated with ADHD different between males and females?**

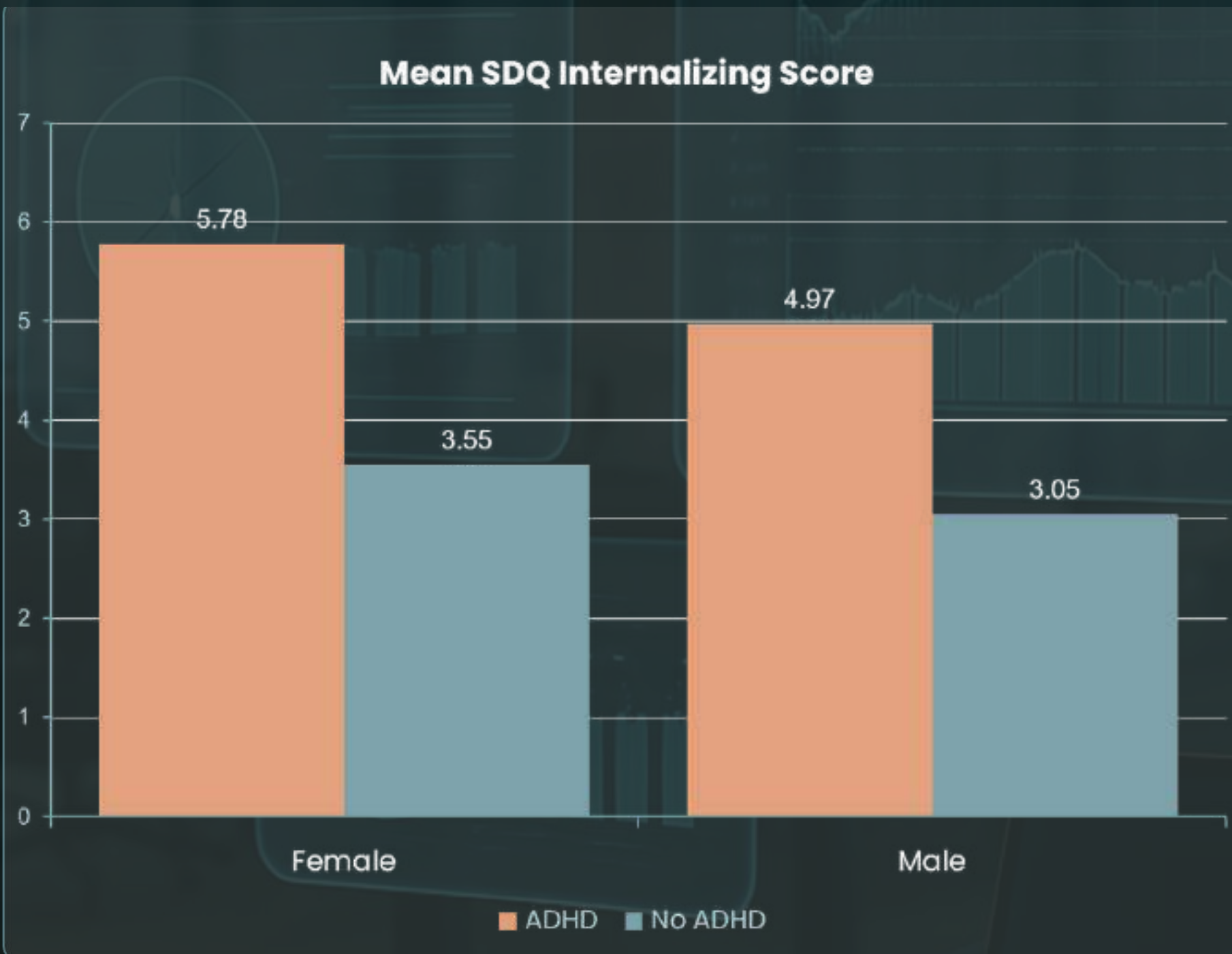
Behaviour + Brain Connectivity, Compared by Sex

Female ADHD is underdiagnosed; girls tend to present with less disruptive, more internalizing symptoms, and the checklists built around boys often miss them.

## BEHAVIOURAL ANALYSIS

# Behaviour looks similar across sexes

SDQ score ~ ADHD + Sex + Sex \* ADHD | FDR corrected



### SIGNIFICANT AFTER FDR CORRECTION

- Internalising  $p < 0.001$
- Emotional Problems  $p < 0.001$
- ProSocial Behaviour  $p < 0.001$

Sex × ADHD interaction: not significant for any SDQ domain (smallest  $p = 0.118$ )

ADHD raises symptom scores the same way in both sexes; girls and boys differ slightly at baseline, but ADHD doesn't affect them differently.

Parenting-practice scores (APQ) showed the same pattern : no sex × ADHD interaction

→ If behavior looks the same, does the brain also?

## ML PIPELINE

# Sex-specific models, built to avoid leakage

*Same features, fully separate models; females and males never share a training fold*



Leakage-safe by design. Imputation, edge ranking, and feature selection are all fit inside the training fold only; never on validation data.

This gives us the stable connectivity edges we'll test formally in a moment →

MODEL PERFORMANCE

# Models distinguished ADHD status reasonably well

AUC across two model types, evaluated separately by sex



READING THE TRADE-OFF

Logistic Regression

Best AUC for males (0.710); the most consistent performer overall

LightGBM

Best AUC for females (0.698); steadiest trade-off between sensitivity and specificity

## STABLE CONNECTIVITY EDGES

# Different connectivity patterns emerged by sex

Edges selected in  $\geq 80\%$  of cross-validation folds: stable candidates for formal testing

### FEMALE — 8 stable edges

- Somatomotor  $\leftrightarrow$  Dorsal Attention
- Visual  $\leftrightarrow$  Somatomotor
- Visual  $\leftrightarrow$  Control
- Limbic  $\leftrightarrow$  Control
- Limbic  $\leftrightarrow$  Somatomotor ( $\times 2$ )
- Control  $\leftrightarrow$  Visual
- Control  $\leftrightarrow$  Limbic

### MALE — 3 stable edges

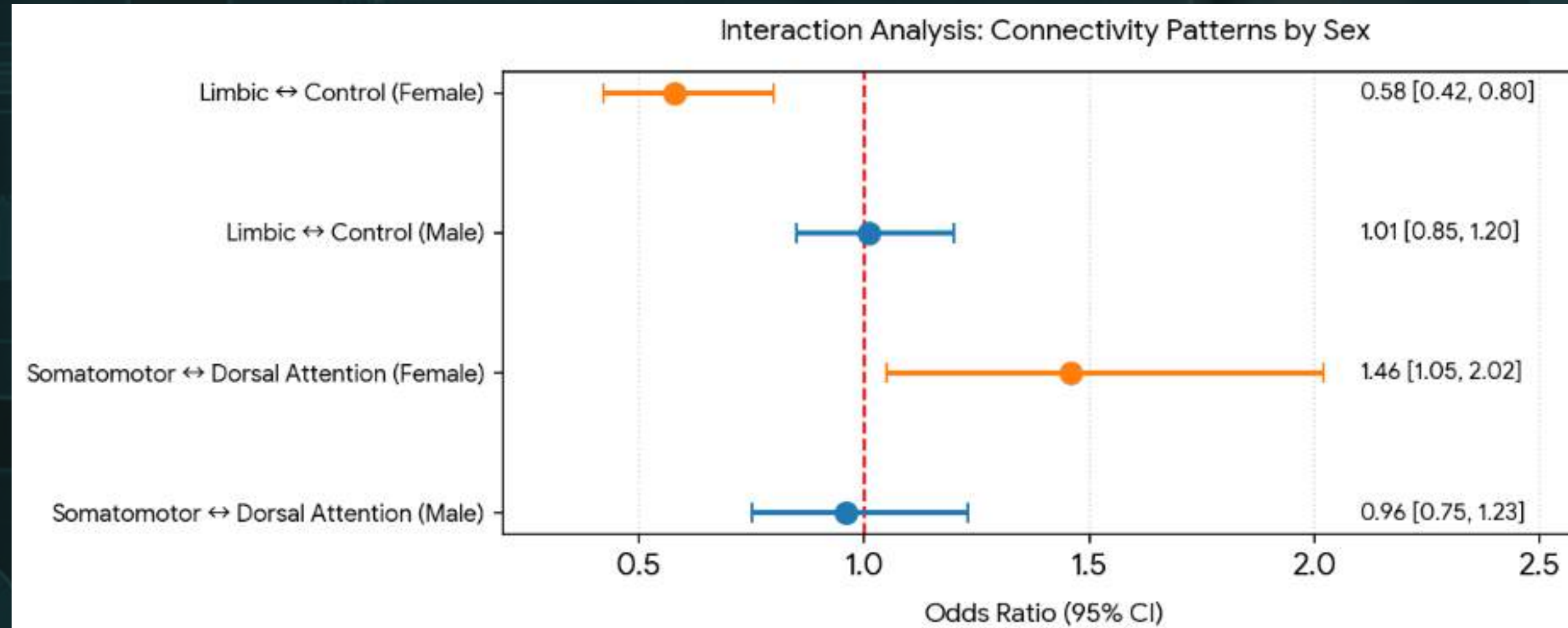
- Limbic  $\leftrightarrow$  Somatomotor
- Default Mode  $\leftrightarrow$  Limbic
- Default Mode  $\leftrightarrow$  Control

***Zero overlap between the female and male stable-edge sets.***

## INTERACTION ANALYSIS

# Two connectivity patterns behave differently by sex

ADHD ~ Edge × Sex + Age + Site | Benjamini–Hochberg FDR correction



### Somatomotor ↔ Dorsal Attention

Higher connectivity → about 46% higher odds of ADHD in females. No meaningful association in males.

### Limbic ↔ Control

Higher connectivity → about 40% lower odds of ADHD in females. No meaningful association in males.

***Selected and tested in the same sample : treat as exploratory and hypothesis-generating, not independently confirmed***

## Conclusion

- Metadata predicts ADHD well; connectivity predicts sex well; combining them doesn't uniformly improve either.
- Behaviourally, ADHD looks the same across sexes. Two connectivity patterns don't; a real signal in females, none in males, confirmed after adjustment.
- Sex-specific signal exists but gets averaged away in pooled models; this is why stratifying by sex mattered.

## Limitations / Next steps

- Edges were selected and tested on the same sample: exploratory, not independently confirmed.
- Female subgroup was smaller (416 vs 797); less power to detect subgroup-specific effects.
- Next step: validate on an independent cohort.

The background is a dark teal color with faint, semi-transparent illustrations of business-related elements. In the top left, there is a 4x4 grid of small white dots. To its right is a line graph showing an upward trend. Further right is a world map. Below the grid is a pie chart and a bar chart. In the bottom right, there is another 4x4 grid of small white dots. The overall theme is business and data.

# Thank you

Any Questions?